

ATTACHMENT -I

SL.NO.	REV.NO	REVISION HISTORY								
1.	E1	Maintenance space around the equipment is revised as “Maintenance space of 1.0 m or as recommended by Equipment OEM”								
2.	E2	Pipe rack free height at road crossing is revised as per construction permit document.								
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MECHANICAL DESIGN CRITERIA (REVISION HISTORY)										
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EPC CONTRACTOR											
 ÇALIK ENERJİ SAN. VE TİC. A.Ş											
CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT (300 MW)											
REVISION HISTORY											
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ATTACHMENT – 1 : GE SUPPLIED EQUIPMENTS THE DESIGN CRITERIA / ASSUMPTIONS

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1. INTRODUCTION

1.1. PURPOSE

The purpose of the document is to furnish the mechanical design criteria for the plant.

2. PROJECT DESCRIPTION

2.1. GENERAL

West African Energy intends to build a combined cycle power plant at Dakar. CALIK ENERJİ” and has received the order from End User to execute the power plant on an Engineering, Procurement, Construction and Commissioning (EPC) basis.

2.2. PROJECT LOCATION

Cap des Biches site is on the edge of the Atlantic Ocean, located in the North Industrial Zone of Dakar, 25 KM from the port of Dakar.

2.3. PLANT CONFIGURATION

The plant configuration comprise of One (1) combined cycle power block in multi shaft configuration. The block consists of two (2) gas turbine and generators (GTGs), two (2) heat recovery steam generators (HRSGs) & one (1) steam turbine and generator (STG), and Balance of Plant (BOP) equipment. The gas turbine is capable of firing tri fuels - Natural gas (Primary fuel), Naphtha (secondary fuel) and Gas Oil (start up fuel) respectively.

The gas turbine will be of Frame 9E.03. The HRSG will be of triple pressure, non-reheat and unfired type. The Steam turbine will be of axial exhaust type with air cooled condenser. The gas turbine, HRSG and steam turbine are supplied by GE.

For more details refer General Design Criteria – SNC -36-VD-IR-CEI-001.

3. MECHANICAL SYSTEMS AND EQUIPMENT

These general mechanical requirements are valid for all power plant equipment, except any additional and/or special requirements mentioned in the particular technical sections specific to the individual equipment/ system/works.

All equipment, materials and the works forming part of this contract shall comply in all respects with any relevant statutory regulations, by-laws or orders currently in force in Senegal.

3.1. MAJOR MECHANICAL SYSTEM

The major mechanical systems other than GE supplied items are listed below:

- a) Exhaust gas system (Bypass stack and damper)
- b) HP/IP/LP Steam System
- c) Auxiliary Steam System
- d) Condensate system including Air Cooled Condenser
- e) Feed Water System
- f) Drain system
- g) Fuel gas system (Natural Gas as primary fuel)
- h) Naphtha System (Secondary fuel)
- i) Gas Oil System (Start-up fuel)
- j) Raw Water System
- k) Service water system
- l) Potable water system
- m) Closed Cooling Water System
- n) Instrument/Service air system
- o) DM water System
- p) Chemical Dosing System
- q) Sampling System
- r) Fire Fighting and Detection System
- s) Heating Ventilation and Air Conditioning System
- t) Waste Water Treatment System
- u) Bulk Gas System
- v) Cranes and handling Equipment
- w) Workshop Equipment

The major mechanical GE supplied items are listed below:

- a) Gas Turbine
- b) Steam Turbine

- c) Heat Recovery Steam Generator
- d) Steam Bypass Valves
- e) CEMS

For GE supplied equipments the design criteria/assumption will be as per Attachment – 1.

4. PIPING

4.1. GENERAL REQUIREMENTS

The engineering of piping shall comply with the conditions specified below.

The piping systems will be designed considering pipe expansion, valve locations, practical pipe supporting and access to all piping accessories for maintenance purposes.

Piping arrangements will provide for the removal and/or maintenance of equipment and equipment internals without removing the piping.

As a minimum 2.2 m headroom will be maintained to the lowest point of all piping components or insulation in walking areas and 1.8 m headroom will be maintained in and around STG deck & ACC duct crossing. Maintenance space of 1.0 m or as recommended by Equipment OEM will be considered for access between and around all equipment and a 100 mm minimum clearance will be provided between piping, including insulation if applied, and any point of adjacent equipment or piping including flanges and/or pipe supports.

E1

Additional clearance will be provided, where required, for pipe movement due to thermal expansion. All valves, instruments, strainers, orifices, valve motor drives and other piping accessories will be provided with adequate space for access and removal of parts for maintenance.

Provision shall be made while preparing piping layout to accommodate all system accessories such as valves/expansion bellows/instrument stubs/instruments/specialties as per P&ID.

All buried pipe in general shall be laid with the top of the pipe minimum 1.0m below the finished ground level.

Bracket supports and other protruding attachments which constitute a hazard to operating personnel will be elevated to a minimum of 2.2 m from the operator's access elevation.

All horizontal runs of outdoor piping will be supported by pipe racks with a slope of 1:500 applicable for steam lines only. The pipe racks will be provided with free height of 4.5 m for road crossing. Steam lines inside building will be laid with a slope of 1:200. The instruments will be positioned to allow easy observation. A 250 mm minimum clear space will be provided from the bottom surface of pipe to trench bottom or finished grade.

E2

All lines will be provided with vent and drain connections at all high and low points, respectively, as per layout constraints with suitability for operation. All piping will be arranged to permit complete drainage when a particular unit or system is shutdown. Condensate & water lines will also be provided with vent & drain lines for hydro test conditions.

Precautions will be taken for fuel oil vents & drains. All vent lines will be terminated 3.0 m or higher above the highest trench, floor drain or sewer.

All Layout and sections drawings and related tags and dimensions will be prepared in AVEVA E3D draw module. Isometrics of all pipe lines irrespective of line size will be provided as a minimum.

3D modelling will be performed according to Calik standards.

This specification does not cover all aspects of design of the piping, but indicates the fundamental principles.

The supplier must submit the classification of piping equipment intended for use in the project for approval by the Purchaser/Engineer.

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4.1.1 Piping codes and specification

Codes and standards applicable for the project are furnished in General Design criteria SNC-36-VD-IR-CEI-001.

Code ASME B-31.1 shall be considered as minimum requirements in relation to the design, fabrication, inspection and testing of the pipes, pipe assemblies, pipe fittings, flanges and components, valves and specialties and any other pressure retaining part in a piping system. EN/BS codes are also accepted, subject to Purchaser review and approval.

As a general rule, metric system shall be used.

The dimensions, tolerances and identifications of all the piping accessories shall confirm with ASME and MSS standards or equivalent.

Piping governed by the ASME code section I shall be designed, fabricated, inspected and tested in accordance with ASME B-31.1 or equivalent.

Section IX of the ASME code shall apply to the welding procedures and performance qualifications.

The system of piping and accessories, to be manufactured from special materials that are not covered by applicable codes and standards, shall be designed according to good engineering and industrial practices for this type of equipment; the supplier shall identify and clearly set out the bases of the manufacturing design it selects, with references to their use.

All piping materials meet the ASTM specifications, their choice will be based on process requirements, the rules of general engineering and experience of their use in similar facilities.

4.2. PIPE SIZING CRITERIA

The maximum allowable velocity of flow in pipes shall be as follows:

Fluid	Min Velocity [m/s]	Max Velocity [m / s]
Fuel gas	15	30
Fire Water	3	5
Air	8	15
Naphtha/Gas oil fuel	1	3
Pump suction	1	1.5 3(BFP Suction)
Pump Discharge	1 2 (BFP Discharge)	3 6 (BFP Discharge)
Drains		0.75
High pressure steam	35	70
IP Steam System	35	70
LP Steam System (Continuous Operation)	35	50
LP Steam System (Non Continuous Operation)	35	70
Auxiliary steam	15	35
Bypass line from the steam bypass v / v to condenser	70	75

Note: For LP Steam, under continuous baseload operations, steam velocity will not exceed 50 m/s.

4.3. PRESSURE DROP

The steam line sizing will be carried out by limiting pressure drop as per contract heat balance for design ambient conditions under NG firing and Naphtha firing. The other balance of piping will be sized to meet the terminal point pressure requirements.

4.4. WALL THICKNESS

The calculation of wall thickness required for pipelines subject to internal and/or external pressure shall be based on the formulae and recommendations as given in ASME B 31.1 / applicable codes. Thickness thus calculated shall then be checked based on the procedure and formula given in local statutory codes as applicable and

the higher value of the two shall be taken for selecting the pipe thickness. Adequate allowances shall be made towards thinning due to bending, weakening at branch connections, threading, commercial tolerances on pipe wall thickness, corrosion and erosion, etc. and the same shall be subject to approval by Purchaser.

4.5. CORROSION AND EROSION

In any case a minimum corrosion allowance of 1.0 mm shall be considered while selecting the thickness. For stainless steel piping corrosion allowance will not be considered.

Pipe handling flashing liquids, shall be selected with extra wall thickness to combat erosion. Downstream of drain control valves shall be alloy steel material complying with ASME A335 Gr P11 / ASME A335 Gr P22.

Much greater wall thickness than called for by strength calculations would be required in order to minimize noise and vibration at certain locations such as downstream sections of high pressure reducing valves. The thickness and lengths of such sections shall be specified.

4.6. PIPING MATERIALS

System	Materials
HP steam piping	ASTM A335 Grade P-22, P91
IP/LP/Aux steam piping	ASTM A106 Gr.B
Condensate piping	ASTM A106 Gr B
Spray Water Piping to Bypass	ASTM A106 Gr B/ ASTM A335 Grade P-22
Feed water piping	ASTM A106 Gr B / ASTM A106 Gr.C
Gas Oil	ASTM A106 Gr.B
Gas Oil (Downstream of Final Filter)	ASTM A 312 TP 304L
Fuel Gas Piping	ASTM A106 Gr.B
Fuel gas piping (downstream of the Coalescing Filter)	ASTM A 312 TP 304
Raw Water	ASTM A106 Gr.B

DM water	ASTM A 312 TP 304
DM Plant (Internal)	CPCV
Naphtha	ASTM A106 Gr.B
Naphtha (Downstream of Final Filter)	ASTM A 312 TP 304L
Fire Water	ASTM A106 Gr.B
Instrument air piping	ASTM A 312 TP 304L
Service air piping	ASTM A106 Gr.B
Service water	ASTM A106 Gr.B
Potable water	ASTM A 53 Gr.B with hot dip galvanised
Bulk gas system	ASTM A106 Gr.B
Waste water from various area	ASTM A106 Gr.B
Chemical Feed System	ASTM A 312 TP 316
Sampling System	ASTM A 213 TP 316

Flange and pipe fitting materials will be selected based on parent pipe selected for various service intended. However all materials will conform to ASTM standards.

For more details refer piping classification SNC -36-VBR-TI-CEI-001.

4.7. DESIGN TEMPERATURE AND PRESSURE

The design temperatures shall as per the recommendations of applicable codes, with required tolerance on the temperature instruments mounted on the respective system.

The design pressure of piping will be maximum operating pressure with sufficient margin as per code requirements. For HP steam, IP Steam and LP steam the piping design pressure will be as per HRSG vendor recommendation. The piping downstream of pumps will be designed for pump shut off pressure plus 5% margin. The piping, which is exposed to vacuum, will be designed for full vacuum.

4.8. PIPE SUPPORTS AND EXPANSION

Pipe supports for HP piping system are decided based on Pipe stress Analysis. Piping Stress analysis using CAESAR II computer software will be performed for the piping system which will fall under any of the following Categories:

- All lines of size, with 120°C fluid temperature.
- All Lines of size 2" (NB 50) and above connected with sensitive rotating equipment.
- Piping connected to Turbines / Compressors / Pumps to satisfy the allowable force & moments.
- Cold piping and small bore piping (less than 50NB) supported using standard piping spans for vertical support from B31.1 / B31.3 (Ref. 1), if unusual conditions or configurations appear to warrant such analysis.

Shoe type supports fabricated from plate materials will be used for pipe size 50DN & below for high temperature pipelines. For pipe size greater than 50DN Trunnion/Stanchion with base plate will be provided for pipe support. Thermal expansion will be taken care by providing spring supports & special supports.

Pipe supports shall be designed according to the layout requirements. Wherever the load variation is limited variable spring hangers and Constant spring hangers are used. In other places other type of supports like rod hangers, clamps, saddle supports etc. are used.

- **Constant Spring Hangers**

The deviation in supporting effect shall be limited to $\pm 6\%$ for constant spring hangers throughout the total travel.

- **Variable Spring Hangers**

The deviation in supporting effect shall be limited to $\pm 25\%$ for variable spring support system.

- **Rod Hangers**

Rod hangers are used in those cases where lateral movement is allowed.

- **Pipe Clamps and Shoe / Saddle supports**

This is used in those cases where restraint support are used. Minimum thickness of pipe clamps shall be 5 mm minimum.

Piping layout is dictated by the relative locations of connecting equipment and the requirements of stress analysis. Adequate number of drain and vents shall be

provided as per the system / layout requirements. Suitable clearances shall be provided with respect to other equipment, structures, pipe to pipe and for ease of access and maintenance. Valves are located and oriented such a way that these are easily accessible for operation and maintenance. No piping shall be routed above GTG/STG. While providing openings in floor/walls for pipe crossing due care shall be taken for the pipe thermal movement and insulation thickness.

4.9. HIGH PRESSURE PIPEWORK

The power cycle piping consists mainly of following.

- a) Main steam piping line from each heat recovery steam generator connected to a common header. Steam piping to the steam turbine will be tapped from this main steam header.
- b) IP steam piping from each heat recovery steam generator connected to a common header. This common header will be feed to STG, The auxiliary steam requirement will be taken from IP steam common header.
- c) This common LP steam piping from each heat recovery steam generator connected to a common header will be feed to STG,
- d) Auxiliary steam and other steam piping as per P&ID.
- e) Boiler feed water pump suction piping.
- f) Boiler feed water pump discharge piping.
- g) Boiler feed water pump recirculation piping.
- h) Condensate extraction pumps suction piping.
- i) Condensate extraction pumps discharge piping.
- j) Boiler blowdown, drains, vents etc.
- k) Steam dumps line.
- l) Spray water lines.

The piping connecting steam turbine, ASME TDP recommendations will be followed.

4.10. LOW PRESSURE PIPEWORK

The Low Pressure (LP) Piping system usually covers piping systems of various services that will deliver to the various areas/systems of the power plant. The piping

systems are basically Balance of Plant (BOP) piping with low temperature and low pressure. These piping are generally routed on suitably sized pipe racks. These pipe racks along with pipe sleepers help in distributing the utilities to their consumption points.

The list of various LP Piping systems are listed below:

- Makeup water piping for GT/STG closed loop circuit.
- Raw water piping (City water)
- Filtered water
- Instrument air
- Service air
- Nitrogen gas
- Fuel gas
- Naphtha (unloading / loading / drains etc.)
- Gas oil (unloading / loading drains etc.)
- Chemical dosing system.
- DM water
- Effluent
- Vents & drains
- Fire fighting piping

4.10.1 General

4.10.2 Compressed air pipe work

The compressed air piping will be provided with moisture traps at every 30 m interval.

4.10.3 Drain and vent Pipe work

Drains shall be provided, at all low points and vents at all high points as per actual layout regardless of whether the same have been shown in the flow diagrams or not. Pipelines shall be sloped towards the drain points.

Besides at low points, tell-tale drains shall be provided at other points as necessary to remove on-line valves and specialties from piping for repair and replacement.

Wherever spray water is introduced into steam piping, at a suitable location downstream, drain pots shall be envisaged along with motor operated drain valves and high and low level switches. The drain valves shall be interlocked with the level switches for automatic operation. For operating temperatures of greater than 400°C, drip leg with drains, operated by a motor operated valve shall be provided. At all other drain locations (operating temperatures less than 400°C) traps shall be provided. All traps shall be provided with strainers, isolating, bypass and free blow valves. Traps shall be thermodynamic type. Traps shall be provided at every 30-40 meters interval. Drain and vent connections on pipelines shall be at-least of NPS 25 mm size.

For pressure and temperature conditions 40 bar & more or temperature greater than 425°C, double isolating valves shall be used on drain and vent lines. For pressure and temperature conditions less than 40 bar or temperature less than 425°C, single isolating valves are considered adequate. Where with single root valve, instrument and root valve are not accessible, double root valve shall be provided with one root valve near the tap off and the second root valve with instrument at accessible point. For Naphtha/ Gas oil drain and vent lines double isolation valves shall be provided irrespective of the pressure and temperature.

4.11. PIPE SPOOL

The pipe spool criteria will be based on Erection philosophy and agreement with package vendor's. In case of spool piping the codal requirement will be followed.

5. VALVES

5.1. GENERAL

All valves shall be suitable for the service conditions under which they are required to operate. The design, construction and choice of materials shall take into account all operational deviations including pressure surge and thermal shock.

Valves having a back-seating facility are preferred, particularly in critical positions.

Unless otherwise specified, the bodies of valves shall be forged or cast steel. Gun metal or cast iron valves shall not be used in any fuel or oil/gas system or any feed-water or similar system where shock loads could result in sudden fracture.

Other than for main turbine stop valves, isolating valves for steam services should normally be of the parallel slide or split wedge gate type.

Gate valves should be such that the bore of the fully open valve is unobstructed by any part of the gate. Where the blade of a butterfly valve requires significant thickening to provide adequate stiffness a through-flow (low loss) design is preferred.

Valves operating under sub-atmospheric conditions shall be with glands and seating rings water sealed or alternatively deep groove gland packing will be provided.

Diaphragm valves shall be used for services carrying chemicals or aggressive fluids, and also for salt water services.

Valves for lubricating oil services shall be flanged, having steel bodies with stainless steel trim.

Any valve which is designed for uni-directional flow shall have an arrow embossed or cast on the valve body clearly indicating the required flow direction.

All valves, unless otherwise approved or specified, shall be of the external rising spindle type. Where desirable to protect the spindle against ingress of dirt, or where the position of the valve may create a hazard to operators when the spindle is extended, suitable spindle covers shall be provided. The spindles and operating gear of all valves for use outdoors shall have weather and dust proof protection.

Any gear or bevel wheels used to transmit motion shall be of cast steel or approved quality cast iron with machine cut teeth. No thrust from the valve shall be transmitted to the extension spindles and valve pedestals shall not be mounted directly on floor plating. Any floor steel work trimmers for supporting pedestals shall be provided by the Supplier.

As far as practicable any maintenance or replacement of wearing parts shall be possible with the valves in situation, but for valves with connecting branches of less than 50 mm nominal bore full access to moving components only will be accepted. The seats of all steam valves shall be accessible for machining for purposes of reclamation or repair.

Eyebolts or similar facilities shall be provided, where necessary, to facilitate the handling of heavy valves or components.

No valve shall be installed with its operating spindle gland below the horizontal centre line so as to avoid dirt entering the gland and to ensure complete drainage.

When valves are required to be locked in position for operation they shall be provided with approved locking system on the spindle (no chain) and with a master key system.

Valves shall have their ends prepared for butt-welding. However, where valves are accepted with flanges by a specific approval by Purchaser, they shall comply with the requirements of the relevant codes and standards for the respective pressure and temperature conditions of the pipe system.

Welding ends should be of the same grade material as the pipe, and be long enough to allow heat treatment during and after welding.

The internal diameter of all valves at the ends adjacent to the pipes shall be the same as the internal diameter of the pipe to which they are joined.

All valves shall be provided with labels or nameplates. These labels shall include in the inscription a valve "name", or abbreviated description of the valve duty, and a unique plant reference number for the valve.

Where a modulating valve in high-pressure service has a full-flow emergency by-pass, the latter shall also be capable of modulating duty.

All high pressure valves of 150 mm bore and above shall be provided with by-passes connected to the valve body. All high pressure gate valves shall be provided with pressure sealed bonnet (breech lock). Maintenance of rupture discs shall be possible during the valve in service.

Drainage and flushing of space between the gate valve wedge or between seat rings shall be provided.

The valve bonnet size shall be such that all surfaces of the valve shall be accessible for repair, lifting, weld surfacing and seat replacement.

Technical Requirements

All globe valves shall be capable of being closed against pressure.

All valves end details shall be as specified in data sheets. In the case of all end-welded valves, the stub ends of the valves shall project from the valve body a sufficient amount to ensure that the welding process will not distort the valve body and internal parts.

Valves for regulating duties shall be of the globe type with tapered plug type disc.

Hardness of disc seating surface shall be minimum of 350 BHN and a minimum differential hardness of 50 BHN shall be provided between disc and seat to prevent galling. This is applicable for stellited seats.

All gate, globe, Y-type and angle valves intended for manual operation and falling under the following categories shall be equipped with a gear operator for ease of operation and to ensure fast and tight closure:

ASME PRESSURE RATING VALVE SIZES FOR WHICH GEAR OPERATOR IS REQUIRED

Class 300 and below	350 mm and larger (for gate and globe valves)
Class 600 and above	200 mm and larger (for gate and globe valves)
Class 300 and above	150 mm and larger (for ball & plug valves)

Class 600 and above	100 mm and larger (for ball & plug valves)
All Class	200 mm and larger (for Butterfly valve)

All gate valves falling in the following categories shall be provided with integral bypass valve. Bypass size shall conform to MSS-SP-45 as a minimum standard unless otherwise specified in data sheet. The bypass valve shall be hand operated unless otherwise specified in data sheet. Pipe for bypass shall be at least Schedule 80 seamless and of a material of the same nominal chemical composition and physical properties as that used for the main line. Orientation of bypass arrangement shall be subject to the approval of Purchaser.

ASME PRESSURE RATING VALVE SIZES FOR WHICH INTEGRAL BYPASS IS REQUIRED

Class 600 and over	150 mm and larger
Class 300 & 150	350 mm and larger

Valves with flexible wedge shall have provision for relieving the pressure in the neck of the valve when the valve is in the closed position.

All valves of ASME pressure rating up to class 600 shall have bolted bonnet construction and class 900 and above shall have bonnet construction of the pressure seal type. Pressure seal valves shall have stainless steel inlay in the gasket area and silver-plated gaskets or other acceptable proven features.

All HP valves of size 50 NB and below shall be of minimum class 600 welded bonnet type irrespective of design requirements.

All valves of ASME pressure rating up to class 900 and higher shall have yoke with anti-friction bearing arrangement.

All valves shall be tested hydrostatically for strength, tightness of seats and tightness of back seating at the pressures indicated in data sheets, in accordance with MSS-SP

61, "Hydrostatic Testing of Steel Valves". Water used for hydrostatic testing of valves with stainless steel components shall not have chloride content exceeding 20 ppm. Clean potable water may be used for testing of all other valves.

All valves except check valves shall be tested for seat tightness by air at a pressure of 6.2 bar (g) on both sides of seat.

All check valves shall also be Hydrotested at 25% of the seat hydro test pressure.

All valves shall be checked for correctness in respect of specified end details as per applicable standards.

All tools needed to lap and seat replacement and repair shall be provided.

All high pressure valves shall be of butt welded construction.

All Motorised valves shall have a motorised bypass valve for pressure equalisation.

5.2. MOTORIZED VALVES

Valves, dampers and similar devices requiring abnormal physical effort to operate or high speed operation, shall be provided with powered actuation. Hand operating equipment shall also be fitted for closing and opening, which is effectively decoupled during power operation.

Electric actuators for on/off duties shall be fitted with travel limiting devices capable of accurate and positive adjustment. Torque limiting or other devices shall be provided to prevent damage to the mechanism in the event of jamming of the driven device.

All powered valve or damper mechanisms shall be provided with accurate indication of the position of the driven device. External limit switches shall be proximity limit type.

Electrically operated devices may be provided with integral or separate switch/ control equipment in accordance with the requirements of the control and instrumentation specification. Provision shall be made for the inclusion of interposing relays and volt free contacts to suit modern control systems, including digital control systems.

Power supply shall be in conjunction to the particular requirements of the control and instrumentation specification.

Direct solenoid operation of valves shall comply with the control and instrumentation section of the specification.

Reversing contactors and limit switches, used to control electric power driven mechanisms, shall be robust and of proven reliability by both type test and commercial operation. Solid state switching may be accepted when temperature conditions permit.

5.3. VALVE MATERIALS

GATE VALVES							
SL NO	SERVICE	SIZE	BODY / BONNET	DISC	STEM	SEAT	END JOINT
1	STEAM TEMPERATURE ABOVE 510° C	<= 50 NB	ASTM A 182 GRADE F91 / A 182 GRADE F22, CL.3	ASTM A 182 GRADE F91 / A 182 GRADE F22, CL.3	13% CHROME	ASTM A 182 GRADE F91 WITH STELLITE HARD FACING / A 182 GRADE F22, CL.3 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 217 GRADE C12A / A 217 GRADE WC9	ASTM A 217 GRADE C12A / A 217 GRADE WC9	13% CHROME	ASTM A 217 GRADE C12A WITH STELLITE HARD FACING / A 217 GRADE WC9 WITH STELLITE HARD FACING	BUTT WELDED
2	STEAM TEMPERATURE ABOVE 420° C AND BELOW 510° C	<= 50 NB	ASTM A 182 GRADE F11, CL.2	ASTM A 182 GRADE F11, CL.2	13% CHROME	ASTM A 182 GRADE F11, CL.2 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 217 GRADE WC6	ASTM A 217 GRADE WC6	13% CHROME	ASTM A 217 GRADE WC6 WITH STELLITE HARD FACING	BUTT WELDED
3	STEAM TEMPERATURE LESS THAN 420° C AND FEED WATER, CONDENSATE, CLOSED	<= 50 NB	ASTM A 105	ASTM A 105 WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	A 216 GR WCB / ASTM A	ASTM A 105 / A 216 GRADE	13% CHROME	ASTM A 105 / A 216 GRADE WCC / A 216 GRADE	BUTT WELDED

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GATE VALVES							
SL NO	SERVICE	SIZE	BODY / BONNET	DISC	STEM	SEAT	END JOINT
	CYCLE COOLING WATER SYSTEM, NITROGEN FILLING		216 GRADE WCC	WCB/ A 216 GRADE WCC		WCB (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	
4	POTABLE WATER	> 50 NB	ASTM A 216 GRADE WCB (GALV)	ASTM A 351 GR.CF8	ASTM A 479 TYPE 410	STELLITE	FLANGED
5	INSTRUMENT AIR	<= 50 NB	ASTM A 182 GRADE F304L	ASTM A 182 GRADE F304L	SS 316	ASTM A 182 GRADE F304L WITH HARD FACING	THREADED
		> 50 NB	ASTM A 182 GRADE F304L / ASTM A 351 GRADE CF8	ASTM A 351 GR.CF8	SS 316	SS 316 WITH STELLITE HARD FACING	FLANGED
6	SERVICE AIR & SERVICE WATER	<= 50 NB	ASTM A 105	ASTM A 105 WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	BUTT WELDED
7	DM WATER DISTRIBUTION	<= 50 NB	ASTM A 182 GRADE F304	ASTM A 182 GRADE F304	SS 316	ASTM A 182 GRADE F304 WITH HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 351 GRADE CF8	ASTM A 351 GRADE CF8	SS 316	SS 316 WITH STELLITE HARD FACING	FLANGED
8	CHEMICAL FEED SYSTEM	<= 50 NB	ASTM A 182 GRADE F316	ASTM A 182 GRADE F316	SS 316	ASTM A 182 GRADE F316 WITH HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 182 GRADE F316 /	ASTM A 351 GRADE CF8M	SS 316	SS 316 WITH STELLITE HARD FACING	FLANGED

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GATE VALVES							
SL NO	SERVICE	SIZE	BODY / BONNET	DISC	STEM	SEAT	END JOINT
			ASTM A 351 GRADE CF8M				
9	EFFLUENT	<= 50 NB	ASTM A 105	ASTM A 105 WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	BUTT WELDED
10	LUBE OIL, WASTE OIL AND CONTROL / HYDRAULIC OIL, (UPSTREAM OF FILTER)	<= 50 NB	ASTM A105	ASTM A 105	13% CHROME	ASTM A 105 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A216 GR WCB	ASTM A216 GR WCB	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	BUTT WELDED
11	LINES, LUBE OIL, WASTE OIL AND CONTROL / HYDRAULIC OIL, (DOWNSTREAM OF FILTER)	<= 50 NB	ASTM A182 F 316	ASTM A 182 GRADE F316	SS316	ASTM A 182 GRADE F316 WITH HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A351 GR CF8	ASTM A 182 GRADE F316	ASTM A 182 GRADE F316	SS 316 WITH STELLITE HARD FACING	BUTT WELDED
12	RAW WATER (CITY WATER)	<= 50 NB	ASTM A 105	ASTM A 105 WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	BUTT WELDED

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GLOBE VALVES							
SL NO	SERVICE	SIZE	BODY / BONNET	DISC	STEM	SEAT	END JOINT
1	STEAM TEMPERATURE ABOVE 510° C	<= 50 NB	ASTM A 182 GRADE F91 / A 182 GRADE F22, CL.3	ASTM A 182 GRADE F91 / A 182 GRADE F22, CL.3	13% CHROME	ASTM A 182 GRADE F91 WITH STELLITE HARD FACING / A 182 GRADE F22, CL.3 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 217 GRADE C12A/ A 217 GRADE WC9	ASTM A 217 GRADE C12A/ A 217 GRADE WC9	13% CHROME	ASTM A 217 GRADE C12A WITH STELLITE HARD FACING / A 217 GRADE WC9 WITH STELLITE HARD FACING	BUTT WELDED
2	STEAM TEMPERATURE ABOVE 420° C AND BELOW 510°C	<= 50 NB	ASTM A 182 GRADE F11, CL.2	ASTM A 182 GRADE F11, CL.2	13% CHROME	ASTM A 182 GRADE F11, CL.2 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 217 GRADE WC6	ASTM A 217 GRADE WC6	13% CHROME	ASTM A 217 GRADE WC6 WITH STELLITE HARD FACING	BUTT WELDED
3	STEAM TEMPERATURE LESS THAN 420°C AND FEED WATER, CONDENSATE, CLOSED CYCLE COOLING WATER SYSTEM, NITROGEN FILLING	<= 50 NB	ASTM A 105	13% CHROME	13% CHROME	ASTM A 216 GRADE WCC	SOCKET WELDED
		> 50 NB	A 216 GR WCB / ASTM A 216 GRADE WCC	ASTM A 105 / A 216 GRADE WCB/ A 216 GRADE WCC	13% CHROME	ASTM A 105 / A 216 GRADE WCC / A 216 GRADE WCB (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	BUTT WELDED
4	POTABLE WATER	<= 50 NB	ASTM A 105 (GALV)	ASTM A 351 GR.CF8	ASTM A 479 TYPE 410	SS 316	THREADED
		> 50 NB	ASTM A 216 GRADE WCB (GALV)	ASTM A 351 GR.CF8	ASTM A 479 TYPE 410 410	SS 316	FLANGED
5	INSTRUMENT AIR	<= 50 NB	ASTM A 182 GRADE F304L	ASTM A 182 GRADE F304L	13% CHROME	ASTM A 182 GRADE F304L WITH HARD FACING	SOCKET WELDED

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GLOBE VALVES

SL NO	SERVICE	SIZE	BODY / BONNET	DISC	STEM	SEAT	END JOINT
6	SERVICE AIR & SERVICE WATER	<= 50 NB	ASTM A 105	13% CHROME WITH STELLITE HARD FACING	13% CHROME	13% CHROME WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	BUTT WELDED
7	DM WATER DISTRIBUTION	<= 50 NB	ASTM A 182 GRADE F304	ASTM A 182 GRADE F304	SS 316	ASTM A 182 GRADE F304 WITH HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 351 GRADE CF8	ASTM A 351 GRADE CF8	SS 316	SS 316 WITH STELLITE HARD FACING	FLANGED
8	CHEMICAL FEED SYSTEM	<= 50 NB	ASTM A 182 GRADE F316	ASTM A 182 GRADE F316	SS 316	ASTM A 182 GRADE F316 WITH HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A 182 GRADE F316 / ASTM A 351 GRADE CF8M	ASTM A 351 GRADE CF8M	SS 316	SS 316 WITH STELLITE HARD FACING	FLANGED
9	EFFLUENT	<= 50 NB	ASTM A 105	13% CHROME	13% CHROME	13% CHROME	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	BUTT WELDED
10	NATURAL GAS LUBE OIL, WASTE OIL AND CONTROL / HYDRAULIC OIL , (UPSTREAM OF FILTER)	<= 50 NB	ASTM A 105	13% CHROME	13% CHROME	13% CHROME	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	BUTT WELDED

GLOBE VALVES

SL NO	SERVICE	SIZE	BODY / BONNET	DISC	STEM	SEAT	END JOINT
				FACING			
11	LINES, LUBE OIL, WASTE OIL AND CONTROL / HYDRAULIC OIL, (DOWNSTREAM OF FILTER)	<= 50 NB	ASTM A182 F 316	ASTM A 182 GRADE F316	SS316	ASTM A 182 GRADE F316 WITH HARD FACING	SOCKET WELDED
		> 50 NB	ASTM A351 GR CF8	ASTM A 182 GRADE F316	ASTM A 182 GRADE F316	SS 316 WITH STELLITE HARD FACING	BUTT WELDED
12	RAW WATER (CITY WATER)	<= 50 NB	ASTM A 105	13% CHROME	13% CHROME	13% CHROME	SOCKET WELDED
		> 50 NB	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB WITH STELLITE HARD FACING	BUTT WELDED

CHECK VALVES

SL NO	SERVICE	SIZE	TYPE	BODY	DISC	SEAT RING	END JOINT
1	STEAM TEMPERATURE ABOVE 510° C	<= 50 NB	SWING CHECK	ASTM A 182 GRADE F91 / GRADE F22, CL.3	ASTM A 182 GRADE F91 / GRADE F22, CL.3	ASTM A 182 GRADE F91 WITH STELLITE HARD FACING / ASTM A 182 GRADE F22, CL.3 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 217 GRADE C12A / ASTM A 217 GRADE WC9	ASTM A 217 GRADE C12A / ASTM A 217 GRADE WC9	ASTM A 217 GRADE C12A WITH STELLITE HARD FACING / ASTM A 217 GRADE WC9 WITH STELLITE HARD FACING	BUTT WELDED
2	STEAM TEMPERATURE ABOVE 420° C AND BELOW 510 °C	<= 50 NB	SWING CHECK	ASTM A 182 GRADE F11, CL.2	ASTM A 182 GRADE F11, CL.2	ASTM A 182 GRADE F11, CL.2 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 217 GRADE WC6	ASTM A 217 GRADE WC6	ASTM A 217 GRADE WC6 WITH STELLITE	BUTT WELDED

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CHECK VALVES							
SL NO	SERVICE	SIZE	TYPE	BODY	DISC	SEAT RING	END JOINT
3	STEAM TEMPERATURE LESS THAN 420° C AND FEED WATER	<= 50 NB	SWING CHECK	ASTM A 105	13% CHROME	HARD FACING 13% CHROME (WITH STELLITE HARD FACING FOR GRATER THAN 600 CLASS)	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 216 GRADE WCC / ASTM A 216 GRADE WCC	13% CHROME	ASTM A 105 / ASTM A 216 GRADE WCB / ASTM A 216 GRADE WCC	BUTT WELDED
4	CONDENSATE, CLOSED CYCLE COOLING WATER, NITROGEN FILLING	<= 50 NB	SWING OR PISTON	ASTM A 105/ A 182 GRADE F22, CL.3	13% CHROME / A 182 GRADE F22, CL.3	13% CHROME A 182 GRADE F22, CL.3 WITH STELLITE HARD FACING	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 216 GRADE WCB// A 182 GRADE F22, CL.3	ASTM A 105 / ASTM A 216 GRADE WCB// A 182 GRADE F22, CL.3	ASTM A 105 / ASTM A 216 GRADE WCB/ A 182 GRADE F22, CL.3 WITH STELLITE HARD FACING	BUTT WELDED
5	POTABLE WATER	<= 50 NB	SWING OR PISTON	ASTM A 216 GRADE WCB (GALV)	ASTM A 351 GR.CF8	STELLITE	FLANGED
6	INSTRUMENT AIR	<= 50 NB	SWING OR PISTON	ASTM A 182 GRADE F304L	ASTM A 182 GRADE F304L	ASTM A 182 GRADE F304L WITH HARD FACING	SOCKET WELDED
7	SERVICE AIR, SERVICE WATER, LUBE OIL RETURN HEADER AND WASTE OIL	<= 50 NB	SWING OR PISTON	ASTM A 105	13% CHROME	13% CHROME (WITH STELLITE HARD FACING GRATER THAN 600 CLASS)	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB	BUTT WELDED
8	DM WATER DISTRIBUTION	<= 50 NB	SWING OR PISTON	A 182 F 304	A 182 F 304	ASTM A 182 GRADE F304 WITH HARD FACING	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 182 GRADE F304 / ASTM A 351 GRADE CF8	ASTM A 351 GRADE CF8	SS 316 WITH STELLITE HARD FACING	FLANGED

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CHECK VALVES							
SL NO	SERVICE	SIZE	TYPE	BODY	DISC	SEAT RING	END JOINT
9	CHEMICAL DOSING	<= 50 NB	SWING OR PISTON	ASTM A 182 GRADE F316	ASTM A 182 GRADE F316	ASTM A 182 GRADE F316 WITH HARD FACING	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 182 GRADE F316 / ASTM A 351 GRADE CF8M	ASTM A 351 GRADE CF8M	SS 316 WITH STELLITE HARD FACING	FLANGED
10	EFFLUENT	<= 50 NB	SWING OR PISTON	ASTM A 105	13% CHROME	13% CHROME	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB	ASTM A 105 / ASTM A 216 GRADE WCB	BUTT WELDED
11	NAPHTHA & GASOIL	<= 50 NB	SWING OR PISTON	ASTM A 105 / ASTM A 182 GRADE F304L	13% CHROME / ASTM A 182 GRADE F304L	13% CHROME / ASTM A 182 GRADE F304L WITH HARD FACING	SOCKET WELDED
		> 50 NB	SWING CHECK	ASTM A 216 GRADE WCB / ASTM A 351 GRADE CF8	ASTM A 105 / ASTM A 216 GRADE WCB /	ASTM A 105 / ASTM A 216 GRADE WCB	FLANGED / BUTT WELDED

BALL VALVES							
SL NO	SERVICE	SIZE	TYPE	BODY	SEAT	BALL	END JOINT
1	NITROGEN FILLING, FUEL GAS, NAPHTHA, GASOIL, INSTRUMENT AIR, SERVICE AIR	<= 50 NB	SINGLE/TWO PIECE BODY	ASTM A 105 / ASTM A 216 GRADE WCB / ASTM A 182 GRADE F304L	SOFT : PTFE / METAL STAINLESS STEEL	SS316	SOCKET WELDED
		> 50 NB	TWO/THREE PIECE BODY	ASTM A 105 / ASTM A 216 GRADE WCB / ASTM A 182 GRADE F304L / ASTM A 351 GRADE CF8	SOFT : PTFE / METAL STAINLESS STEEL / TEFLO N	SS316	FLANGED

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BUTTERFLY VALVES							
SL NO	SERVICE	SIZE	TYPE	BODY / DISC	SHAFT	SEAT RINGS	END JOINT
1	CLOSED CIRCUIT COOLING WATER SYSTEM / PLANT WATER SYSTEM	150NB & ABOVE	WAFFER LUGGED	ASTM A 105 / ASTM A 216 GRADE WCB	A182 GR.304	EPT/BUNA-N/ NEOPRENE	FLANGED

5.4. SAFETY AND THERMAL RELIEF VALVES

Pressure relief valves shall be of the spring-loaded type unless otherwise specified, or unless the Purchaser has given written permission for the use of a different type of valve actuation.

Spring-loaded valves shall be fitted with easing gear which shall not be extended.

Where necessary relief valves shall be supplied with vent pipes, silencers and drains and with gagging equipment for hydraulic testing following erection and in the long-term maintenance of the plant.

All major safety relief valves shall be designed to adjust lift and reset pressure.

5.5. PRESSURE REDUCING CONTROL VALVES

Reducing valves shall be of approved type and shall be perfectly stable, quiet and vibration less in operation when pressure-reducing at any throughput up to the maximum flow and shall be suitable for continuous use at the operating conditions.

Where electrical or hydraulic valve operation is included the operating mechanism shall be remote controlled. On failure of the operating mechanism the valve shall neither close nor open automatically but shall be readily adjustable by hand.

A pressure relief valve shall be provided on the outlet side of each reducing valve, capable of exhausting to atmosphere the maximum discharge of the reducing valve without undue build-up of pressure on the lower pressure side. The discharge from the

relief valve shall be led to safe approved target. Pressure gauges and thermometers shall be provided upstream and downstream of the reducing valve.

5.6. MAIN STEAM STOP AND NON RETURN VALVES

All non-return valves shall be provided with means of draining and depressuring the space between the valve and its adjacent isolating valve. The design shall preclude the possibility of the valve jamming in the open position; the effect of solid particles settling-out in dead spaces within the valve should be recognised in this respect.

Unless stated to the contrary, non-return valves should be rapid- closing. Power-assisted closing is acceptable where necessary for rapid shut-off. For large non-return valves on air and water services, consideration shall be given to the consequences of valve slam and the possible requirements for a hydraulic snubber to minimise slam effects.

5.7. ROOT VALVES

All drains and vents design, size and type shall be DN 25. All internal parts shall be constructed from approved materials and shall be renewable. Valve and steam trap bodies and covers shall be of cast or forged steel and shall be suitable for operation at the maximum working pressure and temperature of the steam piping to which they are connected.

Drain traps shall be provided with inlet and discharge valves and hand-controlled bypasses and when discharging into a closed header they shall also be provided with discharge non-return valves.

6. PUMPS

Pumps should be of the centrifugal type unless technical or strong economic reasons dictate that a positive displacement pump, either rotary or reciprocating, is more appropriate.

All pumps shall be designed for continuous operation unless otherwise mentioned.

If several pumps are used for the same pumping work, these shall be of the same design i.e. interchangeable. Standard pumps with suitable characteristics shall be used wherever practical. Only proven makes and models shall be supplied. All accessories and the overall design of pump sets shall be such that they are suitable for automation as planned for the relevant systems.

6.1. GENERAL REQUIREMENTS OF CENTRIFUGAL PUMPS

Maintenance time should be minimized by the selection of appropriate designs of pump, preferably those that leave casing to piping joints intact, i.e. horizontal split-casing types for large centrifugal pumps and horizontal, back pull-out types for smaller units. Where a significant saving in floor space or improvement of layout can be shown, a vertical pump may be acceptable.

Generally fabricated components for pump parts such as casing, diffuser, impeller, shaft, etc. are not acceptable.

The rating of the driver and the pump's NPSH requirements shall be such that the pump is capable of a flow rate 10% in excess of the maximum flow rate required. Pumps with constant speed drivers shall be capable of at-least 5% (normally 5% on head for clean fluids, 10% for solids- bearing fluids) head increase at rated condition and pump rated speed by replacing with a new impeller or impellers. Offered impeller shall in no case be less than the minimum diameter impeller. The impeller selected shall be smaller in diameter than the maximum size that can be accommodated in the selected size of casing and shall have high efficiency and low NPSH requirement. There shall be an adequate margin on NPSH, normally greater than 1 m for values of NPSH (required) between 2 m and 12 m with flow rates up to the stated maximum.

The pumps for parallel operation shall have characteristics suitable for capacity sharing. The shut off head in such cases shall not be less than 110% of the rated head and not more than 120% of the rated head in horizontal pumps and 140% in the case of vertical turbine pumps.

All pump-motor units shall be installed on substantial base plates, with isolating valves, a discharge non-return valve, air release valves (for both pump casing and seal chamber), drain valves, suction and discharge pressure gauges, unless otherwise stated. The supplier should review the minimum flow rates and supply a leak-off system, if necessary, in which case all minimum flow detection devices, diversion valves and pipework shall be supplied.

Base frame dimensions shall be designed considering that pump accessories and connections (battery limits) are arranged within the outside appearances.

Standard pump local instrumentation (for example local indicator for: pressure, temperature indication, etc.) shall be arranged easy visible on a common local board.

All mating flanges shall be doweled.

Stainless steel shims only shall be used. No corrodible material for embedded supports shall be used.

6.2. TECHNICAL REQUIREMENTS OF CENTRIFUGAL PUMPS

In selecting bearing type, account should be taken of possible vibration damage to the bearing surfaces while the machine is stationary. Where sleeve bearings are used, bearing shells shall be replaceable without removing pump or motor shaft. Bearing housings shall be effectively sealed against loss of lubricant and ingress of fluid or dirt.

All oil lubricated bearings shall have housings fitted with means of drainage, visual oil level indicator and capped filler or a constant level oiler which is accessible whilst the unit is operating. The rise in bearing grease / oil temperature with continuous running of the pump shall be within the allowable limit which shall not exceed 20° C for grease and 30°C for oil lubricated bearing above ambient Grease lubricated bearings shall be connected to readily accessible greasing points either singly or at a battery plate. Lubrication should be by means of a pressure gun.

Where rolling contact bearings are used, they shall meet minimum L-10 rating life of either 25000 hours with continuous operation at rated conditions or 16000 hours at maximum axial and radial loads and rated speed.

Replaceable case wear rings shall be fitted where diameters exceed 150 mm and replaceable impeller neck rings where diameters exceed 250 mm. All wear rings shall be a press fit on or in their respective component and positively locked against rotation.

Casing design pressure shall be at least 1.1 times the pump shut-valve pressure with the pump operating at maximum possible inlet pressure. The pump casing and baseplate shall be capable of withstanding all hydraulic forces, together with externally applied pipe moments and loads and the supplier shall state the pump's pipe load limitations.

The pump unit and all removable components greater than 10 kg in weight shall be fitted with suitable lifting lugs and eyes to facilitate all erection and maintenance tasks. Any special lifting tackle shall be supplied.

Suitable means, such as jacking screws, shall be provided to aid the separation of pump components; these shall be arranged so that no damage can be done to mating or sealing faces by their use.

The method of fitting the impeller to the shaft shall adequately centre the component but allow its removal without risk of damage to either component. Torque shall be transmitted to the impeller by keying, other than in the case of a multistage vertical pump for which a collect drive may be acceptable. The main impeller nut or screw shall not have its thread exposed to the pumped fluid and must either be locked or maintained tight by virtue of the normal direction of rotation.

The pump shafts shall be designed such that the first critical speed of the pump is at least 20% higher than the maximum speed encountered in service.

All rotating components shall be dynamically balanced to grade G2.5 of ISO standard 1940. The complete rotating assembly shall also be balanced to Grade G6.3, other than (a) on vertical spindle, multistage pumps or (b) where pump speed is less than 1800 rpm and capacity less than 50 m³/h.

Where two pumps are connected in parallel, the units shall be capable of reverse rotation under full pump head without sustaining any damage, provided there is no non return valve in the discharge valve.

When the pump is operating above its normal flow range, the vibration intensity measured at the bearing housings shall not exceed a level of 4.5 as defined within ISO standard 2372. For small machines (i.e., less than 50 kW driver rating) vibration intensity shall not exceed a level of 2.8.

6.3. SHAFTS, SEALS AND COUPLINGS

Air release valves shall be fitted at suitable points on the pump casing.

Packed glands, if not otherwise specified, shall be used for pumps handling cold water and other non-hazardous liquids. Clean water sealing arrangements shall be provided for gland packing of pumps handling brine, seawater or similar.

Mechanical seals shall be preferably of a pressure-balanced type. The design of the unit should allow the mechanical seal to be replaced without disturbing the pump or driver if practicable.

Mechanical seal shall be fitted over a sleeve on shaft. The mechanical seal shall be of the split type to facilitate quick removal/replacement without any disassembly of other parts of the pump.

Any leakage at the gland shall be intercepted by a flinger.

All pump glands shall be provided with trays and guards to catch leakage. Drainage piping shall be fitted to an approved location.

Other than on small pumps (i.e. with driver less than 50 kW) auxiliary pipework for seat harnesses, drains, etc. shall be not less than 20 mm bore.

Couplings shall be dynamically balanced to grade 6.3 ISO standard 1940 where speed exceeds 1800 rpm or shaft diameter exceeds 51 mm. Dynamically balanced couplings shall be match marked. Where a spacer coupling is used, the spacer shall similarly comply.

All coupling and any intermediate shafts shall be fully guarded.

Pumps handling treated water shall be provided with mechanical seals only. No gland packing is acceptable

6.4. PUMP MATERIALS

Materials of constructions shall be selected with due regard to the corrosive nature of the pumped fluid and risk of corrosion, cavitation and metal-to-metal galling that can occur within a pump in all its modes of operation. It is the supplier's responsibility to select materials that will provide an adequate service life.

If the pump impeller and impeller shaft are of different material, any portion of the impeller shaft which could be in contact with the water must be encased in a protection sleeve of the same material as the impeller. All materials shall be selected with particular regard for the risk of electrochemical corrosion.

Cast iron is not acceptable for pump cases where the duty involves shock loads or where service stresses exceed 10% of the Ultimate Tensile Strength in tension or 15 % in bending. This restriction does not apply to castings of spheroidal graphite structure with an elongation greater than 10%.

The sleeve bearings of the seawater pumps shall be of fibre reinforced silicon carbide (ceramic). Make of the same is subject of approval.

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6.5. GENERAL REQUIREMENT OF SUMP PUMP

Sump pumps shall be used to dewater various sump pits in various plant areas.

Pump shall be permanently fixed or portable in the pits/sumps. If the pit depth is shallow, vertical top mounted sump pumps shall be provided and in deep pits self priming drainage pumps horizontal type) at floor level or alternatively submersible type pumps may be provided.

Each pit/sump shall be provided with two numbers (2 x 100% Capacity) of respective type pumps so that the entire pit is evacuated within 15-20 minutes and the operation of the pumps shall be interlocked through level measurement devices to be installed in the pit/sump so that the pumps shall start automatically and empty the pit.

Each sump pump shall be equipped and coupled with a drive motor, with rating so selected as to have at least 10% margin over the maximum power required by the pump, throughout its range of operation.

The submersible pump-motor type, single stage and non-clog design and shall be portable type.

The impeller shall be open/semi-open non clog type, cast in one piece and specially designed to pass large solids or unscreened liquids. The clearance between stationary and moving parts should be such as to allow sustained performance without exclusive maintenance. The pumps shall be capable of passing fibrous material like cotton waste, jutes, etc.

The Shaft size selected shall be such that critical speed is at least 20% away from the operating speed and the runaway speed.

7. COMPRESSOR

Compressors are generally divided into three major types, dynamic, positive displacement and thermal.

The choice of the type of compressor, whether axial, centrifugal, reciprocating or rotary, depends primarily on the required flow to be compressed, the density of the gas in conjunction with the total head.

1. Axial compressors:

Axial compressors can handle large volume flow and are more efficient than centrifugal compressors. However, centrifugals are less vulnerable and hence more reliable, have wider operating ranges and are less susceptible to fouling. Axial compressors shall be considered only for air, sweet natural gas or non-corrosive gases.

2. Centrifugal compressors

The centrifugal compressor can handle the required flow with a good efficiency, and continuously operate for long periods. If the flow at discharge conditions is 300 m³/h or more, then centrifugal compressor will be used. The centrifugal compressor will be designed accordance with API 617 with standard manufacturer deviation.

3. Reciprocating compressors Where the required flow is too small for a centrifugal compressor, or where the required head is so high that an undesirably large number of stages would be necessary, then generally the choice should be a reciprocating compressor. As a reciprocating compressor cannot fulfill the minimum requirement of continuous uninterrupted operation, due to fairly high maintenance requirements, a full capacity spare shall be provided as general rule for reciprocating compressors in critical services.

The air compressor will be oil free screw type and fuel gas compressor will be of internal geared centrifugal compressor.

Compressors shall be guaranteed for head, capacity, and satisfactory performance at all specified operating points and further shall be guaranteed for power at the rated point.

The volume capacity at the surge point shall not exceed the specified percentage of normal capacity at normal speed, and normal (unthrottled) suction conditions. The rise in pressure ratio from normal capacity to the surge point at normal speed shall not be less than that specified. The head developed at 115% of normal capacity at normal speed shall be not less than approximately 85% of the head developed at the normal operating point.

The head-capacity characteristic curve shall rise continuously from the rated point to the predicted surge. The compressor, without the use of a bypass, shall be suitable for continuous operation at any capacity at least 10 percent greater than the predicted approximate surge capacity shown in the proposal.

For variable speed compressors, the head and capacity shall be guaranteed with the understanding that the power may vary $\pm 4\%$. e. For constant-speed compressors, the specified capacity shall be guaranteed with the understanding that the head shall be within $\pm 5\%$ and - 0% of that specified; the power shall not exceed stated power by more than 4%. These tolerances are not additive.

8. TANKS AND PRESSURE VESSELS

8.1. GENERAL

The design of the Naphtha and Gas oil tanks shall comply in all respects with the requirements of API 650, latest edition and as supplemented by this specification. Water tanks shall be designed in accordance AWWA D100.

The tanks shall be suitably designed for the storage and operation of the particular liquids under the prevailing ambient conditions.

The minimum required thickness shall exclude mill tolerances, fabrication allowances and corrosion allowances in accordance with this specification.

The fabrication drawing shall indicate the orientation, the location, the preparation and the welding sequence of all welded joints on the shop drawings submitted for approval. No weld joint is allowed behind any plate or construction, which prevents thorough inspection of the weld.

The tanks shall be supplied with all internals as required. They shall be adequately supported and shall be removable in pieces through a manhole. The design of internal fittings shall ensure that they are proof against loosening by vibration etc. Internals shall be such that they are easily dismantled for maintenance of the tank without the need of any hot work.

All bolts used in the tank, at nozzle connections or used as anchor bolts shall be hot dip galvanised.

The minimum diameter for the bolting shall be 12 mm for mild steel and 10 mm for stainless steel. All bolts shall have ISO-metric screw threads; UNC type threads are also acceptable, subject to approval of EPC Contractor.

After completing the hydro test, oil storage tanks shall be calibrated for dipping volume measurement with the approved agency and relevant documents shall be submitted the Purchaser. All gaskets shall be free of asbestos.

Tank shall be protected where applicable by foam injection at top and also by spray water cooling of roof and shell.

8.2. MATERIALS

Stainless Steel Tanks Materials

For all plates, the manufacturer shall obtain mill certificates from the steel maker; the certificates shall be made available to the Purchaser. The mill certificates of all plates shall indicate conformity with the material specifications. Mill certificates for all plates with a specified minimum yield strength exceeding 240 N/mm^2 shall be authenticated by an approved authority.

To prevent electrochemical corrosion, parts e.g. stairs, etc., which may not be of stainless steel, shall be connected to the tank in accordance with the relevant accepted practice upon receiving approval from the Purchaser. All work procedures including verification of type of materials prior to the construction of the tank is subject to the approval by the Purchaser. All gaskets, sealing, etc. shall be free of chlorine. When constructing the tank, the stainless steel material shall not come into contact with carbon steel. All stainless steel material shall be protected from carbon steel contamination.

Carbon Steel Tanks Materials

The plates used to form the shells shall be of the grades specified by the API 650. The use of high tensile steel shall only be considered as an alternative. The material listed below, may to the discretion of the Contractor, be replaced by materials according to other standards, provided their equivalence is proved. Bottom surround plates shall be of the same grade as the lower shell sections.

For all plates, the manufacturer shall obtain mill certificates from the steel maker; the certificates shall be made available to the Purchaser. The mill certificates of all plates shall indicate conformity with the material specifications. Mill certificates for all plates with a specified minimum yield strength exceeding 240 N/mm² shall be authenticated by an approved authority.

Bottom and roof plates shall be according to ASTM A 283 Specification Grade C.

The angle irons, and top and side reinforcing segments shall be ASTM A.7. The remaining sectional steel shall also be of the grade recommended by this standard.

Rolled and welded plate nozzle necks as well as tube stiffeners shall be made of steel of the same grade as the plate onto which these fittings are connected.

The pipe studs shall be made of seamless drawn tube as per API 5L, Grade A or B. Forged steel fitting and tube flanges shall be ASTM A 181, Grade I.

Bolts on nozzle flanges, manholes and blind flanges shall be of a grade corresponding to ASTM A.193-B7, A.194-2H. Internal bolts shall be of A 307 B or equivalent.

Gaskets used for seals on piping and fillings shall be made of hydrocarbon resisting compressed klingerite or equivalent type, 1.5 mm gauge up to 450 mm and 3 mm above.

General Note: Material can be selected as EN corresponds for specified ASTM materials.

Construction Features

General

The tanks shall be of the vertical straight cylindrical type constructed of stainless steel plates or carbon steel plates and shall be manufactured and tested in accordance with AWWA D100 - Welded Steel Tanks for Water Storage and API 650 - Welded Steel Tanks for Oil Storage whereby the minimum requirements of API 650, latest edition shall be fulfilled. All welding and welding inspection shall be executed according to API 650, latest edition. The shell joints shall be inspected by radiographic methods. The bottom shall be tested by the crack detection method and by vacuum testing. All work procedures including verification of type of materials prior to the construction of the tank is subject to the approval by the Purchaser/Employer.

Each tank shall be equipped with all connecting branches including those for the following connecting pipes and instrumentation:

- Filling pipe
- Overflow pipe
- Vent pipe
- Draining pipe
- Suction pipe(s) to all pumps
- Return pipes
- Two spare nozzles at the bottom of the tank with blind flanges

- The instrumentation and control equipment
- Level indication
- Sample taking

Protection against Lightning and Static Electricity

Special protection against lightning is also required. The earthing system to be provided for each tank shall have a resistance of 0.5 Ohm at the maximum. The material shall be galvanised copper, no aluminium shall be supplied. The area of copper strips shall be 100-mm² minimum. Earthing poles have to reach down to the ground water level. The Contractor shall explain in detail in which manner possible corrosion of the earthing strips will be prevented (See also applicable NFPA Standard).

All parts inside the fuel oil tanks must correspond to the electrical potential of the shell.

All metal parts inside the fuel oil tanks must be spark-proof.

All nozzles must be sized to keep flow velocity lower than 0.5 m/s.

9. HEAT EXCHANGERS

The heat exchangers shall be designed, manufactured and erected in accordance with the latest revision of international or national codes e.g. TEMA, ASME, BS, manufacturer standard.

Unless otherwise specified, all heat exchangers shall be designed for continuous operation. They must be capable of withstanding conditions resulting from rapidly changing loads without damage and without reduction of the design service life.

It must be possible to install and remove the heat exchangers without undue difficulty. Lifting lugs and eyes and other special tackle shall be provided to permit easy handling of the heat exchangers and their components. Heavy covers shall as far as possible be provided with opening devices, e.g. hinges or davits.

Fixed tube sheets may be used with approval for non-corrosive streams provided that any fouling of shell can easily be removed by hot water solvent and chemical cleaning method. During the design of fixed tube sheet heat exchangers consideration must be given to the stresses induced at operating, start up, shut down and other upset conditions. Heat exchanger with expansion bellows should be avoided.

Heat exchangers with straight tube shall be installed with one end fixed and the other end with floating along with sliding gaskets to avoid thermal expansion.

Unless otherwise specified, all heat exchanger tubes and shells must be designed to withstand 1.2 times the shut off pressure of the relevant pump, or 1.2 times the maximum positive operating pressure, as applicable. The design shall be proof against full vacuum. The test pressure must be 1.5 times the design pressure.

Where reference is made in the various sections to heat exchanger materials, these shall be understood as a minimum requirement and do not relieve the Supplier of the responsibility for selecting the correct material. The use of cast iron is not permissible. Forged tube endplates shall be formed from a single piece of material.

Unless otherwise specified, the heat exchanger tubes shall be seamless tubes. The heat transfer area shall be justified with calculation for the heat load for which the heat exchanger is designed.

The design of the heat exchangers shall be so laid out that groups of tubes or individual tubes can be readily removed, or in the case of leaks, tubes can be easily plugged. The requisite space for removal must be provided by suitable design. The heat exchangers must be capable of continuous unrestricted operation with up to 10% of tubes plugged, and a corresponding factor of conservatism of at least this amount must be used in the design of the heat transfer areas.

Considerable importance shall be paid to the ease of cleaning the heat exchangers. For cleaning purposes the cooling water connections shall be provided with side-mounted blanked flanged stubs with a nominal bore of 50 mm. Water vessels must be provided

with flanged covers. Connecting stub-pipes and connections for drains shall not be less than 15 mm bore.

Minimum spot radiography shall be carried out for welds.

Where tubes are welded to tube plates the whole shall be stress relieved. Tubes shall be stress relieved after bending if required according to code.

Bolting and gasket materials shall be of high quality appropriate to the duty concerned.

All heat exchangers shall be equipped at their highest and lowest points with an adequate number of air vents, water drains and emptying ports along with isolation ball valves. Where heat exchangers operate at sub-atmospheric pressure, the design and the arrangement of air suction ports shall be so coordinated that the formation of air pockets is not possible.

Where any heat exchanger parts in contact with liquid can be isolated and there is a possibility of a pressure rise within the exchanger for whatever reason, safety valves shall be provided for pressure relief.

Pipes from water drains, air vents and safety valves shall be grouped together where possible, and routed to easily observable points equipped with covered funnels or flash tanks. The corresponding valves and fittings must be clearly designated and easy to operate.

The overall design and conception of the heat exchangers and accessories shall be such that they are suitable for the degree of automation envisaged for the individual system.

10. CORROSION PROTECTION FOR BURIED PIPING

All buried Carbon Steel piping shall be wrapped and coated for corrosion resistance. Pipes to be buried shall be cleaned internally & externally thoroughly. Primer and coal tar enamel / Polymer modified bitumen tape shall be supplied by the same manufacturer and coating application procedure shall be in strict compliance with the manufacturer's instructions.

Cathodic protection will be provided for buried piping.

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11. INSPECTIONS AND TESTS

11.1. GENERAL:

The Inspection and Test Plan, shall be provided by the equipment supplier (Contractor) indicating all inspections and tests to be performed with the specified equipment. The Inspection and Test Plan shall refer to the applicable inspection and test instruction, whether for source, incoming, intermediary, final or site inspections and tests. The Contractor's Inspection and Test requirements shall fulfill the minimum inspection and test requirements. The Purchaser, the licensing authorities, as well as their agents shall have the right to witness all tests and inspections carried out in compliance with the Contractor's Inspection and Test Plan. They shall be permitted access the Contractor's shop, where and while all tests or inspections are being performed.

Shop Tests and Inspections: The shop tests and inspections of the specified equipment shall be carried out according to the Contractor's Inspection and Test Plan, which has been approved by the Purchaser.

Site Tests and Inspections: The site tests and inspections of the specified equipment shall be carried out according to the Contractor's Site Inspection and Test Plan, which has been approved by the Purchaser.

Material or equipment shall not be shipped before the contractor presents a certificate of compliance, signed and dated by the contractor's quality assurance/quality control department. A copy of this document shall be securely attached to the equipment to be shipped. It is important that the equipment arrives at the site with a copy of the certificate of compliance attached.

11.2. THE CONTRACTOR'S RESPONSIBILITIES FOR SUBCONTRACTORS

The Contractor shall identify in purchase documents to its Suppliers, all applicable quality and QA requirements imposed by this specification on the Contractor and shall

ensure compliance thereto. The Contractor has the prime responsibility for subcontractor surveillance and evaluating and monitoring the implementation of the Quality Assurance Programs of its subcontractor.

The Contractor shall submit a copy of its subcontractor Surveillance plans to the Purchaser for review prior to implementation. Upon request, the Contractor shall submit to the Purchaser copies of the reports of the subcontractor surveillance activities that it performs on its Suppliers.

11.3. WITNESS AND HOLD POINTS:

- a. The Purchaser shall have the right to establish witness and hold points. In addition, the Purchaser may establish witness points to ensure resolution of quality problems.
- b. Witness points are defined as predetermined points during installation when installation procedures or activities may be inspected or witnessed for compliance with the Contract, specification, and quality requirements.
- c. Hold points are defined as predetermined points during the installation process when procedures must be inspected and accepted by the Purchaser prior to continuing installation. The Purchaser shall have the right to access the Contractor's and its subcontractor's facilities at any time to verify the progress and quality of the work.

Unless otherwise proven and specifically agreed, welding of dissimilar materials and high alloy materials shall be carried out at shop only.

No welding shall be carried out on cast iron components for repair.

All the heat treatment results shall be recorded on time temperature charts and verified with recommended regimes.

All non-destructive examination shall be performed in accordance with written procedures as per International Standards, The NDT operator shall be qualified as per SNT-TC-IA (of the American Society of non-destructive examination). NDT shall be recorded in a report, which includes details of methods and equipment used, result/evaluation, job data and identification of personnel employed and details of correlation of the test report with the job.

12. MAINTENANCE SPARE PART AND SPECIAL TOOLS

12.1. MAINTENANCE SPARES AND CONSUMABLES

All spares supplied shall be strictly interchangeable with the parts, which they are intended to replace and shall be of the same quality as the original parts. The consumables required for commissioning shall be supplied by respective equipment supplier.

12.2. OVERHAUL SPARES AND CONSUMABLES

The overhaul spares and required consumables list will be provided by respective equipment supplier.

12.3. STRATEGIC BREAKDOWN SPARES

Capital spares, items that would cause more than 90 days downtime if not stocked shall be indicated by equipment supplier. The spares required routine maintenance between overhauls will be provided.

12.4. SPECIAL TOOLS

Any special tools and test equipment required to adjust, dismantle, reassemble or maintain any equipment supplied shall be provided.

12.5. START-UP/COMMISSIONING SPARES

The required start-up/Commissioning spares for the equipment/system will be supplied along with equipment/system by respective package supplier.